CORROTECH ITK LTD.

Hong Kong

INDUSTRIAL HEAT EXCHANGERS

Shell & Tube | Plate | Spiral | Air Cooled

For Refinery, Petrochemical,
Chemical & Power Generation

www.pipefittings-hg.com

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ABOUT CORROTECH ITK LTD.

CorroTech ITK Ltd.
Hong Kong

is a Hong Kong-based supplier specializing in critical piping components and expansion solutions for demanding industrial applications.

CONTACT INFORMATION

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OUR EXPERTISE IN HEAT EXCHANGERS:

- ✓ Direct partnerships with certified manufacturers (ISO 9001, ASME, PED)
- ✓ Custom engineering per TEMA, ASME, API, EN standards
- ✓ Materials: Carbon Steel, Stainless Steel, Titanium, Nickel Alloys
- ✓ Full documentation: MTC 3.1, EN 10204, calculation reports, P&ID
- ✓ Third-party inspection available (SGS, BV, TUV, Lloyd's)
- ✓ Flexible payment via Hong Kong entities
- ✓ Global logistics: CIF/FOB/DDP worldwide

INDUSTRIES WE SERVE:

- Oil & Gas Refining
- Petrochemical & Chemical Processing
- Power Generation (Thermal, Nuclear, Combined Cycle)
- LNG & Cryogenic Applications
- Marine & Offshore
- Pulp & Paper, Food & Beverage, Pharmaceuticals

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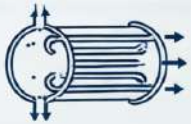
TYPES OF HEAT EXCHANGERS



1. SHELL & TUBE HEAT EXCHANGERS

- Most versatile and widely used type
- High pressure / high temperature capability
- Easy maintenance and cleaning
- Heat transfer area: 1 – 10,000+ m^2
- Design pressure: vacuum – 400 bar
- Design temperature: -196°C to +600°C

2. PLATE HEAT EXCHANGERS (Gasketed / Brazed / Welded)

- Compact design, high efficiency
- Easy to expand capacity (add plates)
- Gasketed: serviceable, flexible
- Brazed/Welded: high pressure, aggressive media
- Heat transfer area: 0.5 – 2,500 m^2
- Design pressure: up to 40 bar (gasketed), 100+ bar (welded)

3. SPIRAL HEAT EXCHANGERS

- Self-cleaning design for fouling services
- Compact footprint, true counter-current flow
- Ideal for slurry, viscous fluids, crystallizing media
- Heat transfer area: 10 – 2,000 m^2
- Design pressure: vacuum – 60 bar

4. AIR COOLED HEAT EXCHANGERS (Fin-Fan)

- Water-free cooling for arid locations
- Forced or induced draft configurations
- Ideal for remote sites, offshore platforms
- Heat duty: up to 100+ MW per unit
- Design temperature: up to 400°C process side

5. SPECIALTY HEAT EXCHANGERS

- Double-pipe / Hairpin: high pressure, small duty
- Plate-Fin: cryogenic, compact, multi-stream
- Scraped-surface: high viscosity, crystallizing
- Falling film evaporators: concentration, distillation

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SHELL & TUBE HEAT EXCHANGERS — DETAILED SPECIFICATIONS

CONFIGURATIONS

	TEMA Type	Description	Application
BEM	Fixed tubesheet	Clean services, low thermal stress	
BEU	U-tube	Thermal expansion, high ΔT	
BEP	Externally sealed floating head	Hazardous fluids	
AES	Internally sealed floating head	High pressure, dirty services	
AEU	Split-ring floating head	Large diameter, easy maintenance	
AKT	Kettle reboiler	Vaporization, distillation	

TUBE SPECIFICATIONS

Parameter	Range / Options
Tube OD	6 mm – 50 mm (1/4" – 2")
Tube Length	1 m – 12 m (custom longer)
Tube Pitch	1.25x – 1.5x OD (triangular/square)
Tube Layout	30°, 45°, 60°, 90°
Passes	1, 2, 4, 6, 8 (custom)
Baffle Type	Segmental, Rod, Helical, Nozzles
Baffle Cut	15% – 45% (optimized for ΔP)

SHELL SPECIFICATIONS

- **Shell ID:** 150 mm – 3,000+ mm (6" – 120+")
- **Shell Type:** E (1-pass), F (2-pass), G (split flow), H (double split), J (divided flow), X (cross flow)
- **Nozzle Orientation:** Top, side, bottom; axial or radial
- **Expansion Joint:** Bellows type for thermal compensation

MATERIALS

Component	Material Options
Tubes	CS, SS 304/316/321/347, Duplex 2205/2507, Titanium Gr.2/5, Inconel 625/825, Hastelloy C-276, Copper-Nickel 90/10, 70/30
Tubesheet	Clad (CS + SS/Ti), Solid alloy, Explosion-bonded
Shell	CS (SA-516), SS, Duplex, clad
Baffles	SS 304/316, CS, non-metallic (PTFE)

PLATE HEAT EXCHANGERS — SPECIFICATIONS

PLATE TYPES & PATTERNS:

Pattern	Characteristics	Application
Herringbone	**(Chevon)** High turbulence, high heat transfer, high ΔP	General service, water, oils
Washboard	Low fouling, gentle flow, low ΔP	Viscous fluids, food, pharma
Mixed	Balanced performance	Multi-service

GASKETED PLATE EXCHANGERS:

- **Plates:** SS 304/316, Titanium Gr.1/2, Nickel 200, Hastelloy C-276
- **Gaskets:** NBR (water/oil), EPDM (steam/hot water), Viton® (chemicals), PTFE (aggressive)
- **Frame:** Carbon Steel epioxy-coated, SS 304/316
- **Heat transfer area:** 0.5 – 2,500 m^2 per unit
- **Design pressure:** up to 25 bar (standard), 40 bar (high-pressure)
- **Design temperature:** -50°C to +180°C (gasket dependent)
- **Advantage:** Easy to clean, expandable, serviceable

BRAZED PLATE EXCHANGERS:

- **Construction:** Copper-brazed or nickel-brazed stainless steel
- **No gaskets** — fully welded plate pack
- Compact, leak-proof, maintenance-free
- **Heat transfer area:** 0.1 – 200 m^2
- **Design pressure:** up to 30 bar (copper), 45 bar (nickel)
- **Design temperature:** -196°C to +230°C
- **Ideal for:** Refrigeration, hydraulics, small process duties

WELDED PLATE EXCHANGERS:

- Fully welded plate cassette (laser/TIG)
- Handles aggressive media, high pressure/temperature
- Heat transfer area: 10 – 2,000 m^2
- Design pressure: up to 100 bar
- Design temperature: -196°C to +400°C
- Ideal for: Chemical, petrochemical, offshore, cryogenic

SPIRAL & AIR COOLED — SPECIFICATIONS

SPIRAL HEAT EXCHANGERS

DESIGN FEATURES:

- Two concentric spiral channels for true counter-current flow
- Self-cleaning action: high velocity prevents fouling
- Compact footprint vs. shell & tube
- Single gasket design (only at periphery)
- Easy access for inspection (cover removal)

SPECIFICATIONS:

	Parameter
Heat Transfer Area	10 – 2,000 m^2
Channel Gap	4 mm – 30 mm (custom)
Design Pressure	Vacuum – 40 bar
Design Temperature	-50°C to +400°C
Materials	SS 304/316, Duplex, Titanium, Hastelloy

IDEAL APPLICATIONS:

✓ Slurry, pulp, wastewater (self-cleaning)
✓ Crystallizing, polymerizing media
✓ High-viscosity fluids (heavy oils, resins)
✓ Condensation with non-condensables

AIR COOLED HEAT EXCHANGERS (FIN-FAN)

CONFICURATIONS:

Type	Description	Application
Forced Draft	Fan below bundle	Easy maintenance
Induced Draft	Fan above bundle	Better air distribution
V-Configuration	Angled bundles	Compact footprint
Steam Condenser	Large surface, vacuum	Power plants

MATERIALS FOR HEAT EXCHANGERS

MEDIA COMPATIBILITY & MATERIAL RECOMMENDATIONS:

Media / Service	Recommended Materials
Cooling Water	CS, SS 304/316, Cu-Ni 90/10
Seawater	Titanium Gr.2, Cu-Ni 70/30, Super Duplex 2507, AL-6XN
Steam / Condensate	CS, SS 304/321, SS 316
Hydrocarbons (oil)	CS, SS 304/316
Sour Service (H_2S)	SS 316, Duplex 2205, Inconel 825
Chlorides	Duplex 2205/2507, Titanium, Hastelloy C-276, AL-6XN
Acids (H_2SO_4, HCl)	Hastelloy C-276, Tantalum-lined, PTFE-lined, Graphite
Caustic (NaOH)	Ni 200, SS 304/316 (dilute)
Cryogenic (LNG, N_2)	SS 304L/316L, Aluminum, 9% Nickel
High Temperature	(>400°C) | SS 321/347, Inconel 600/625, Alloy 800H

CLADDING & LINING OPTIONS:

- Explosion-bonded clad: CS + SS/Ti/Ni alloy (tubesheets, shells)
- Weld overlay: corrosion-resistant layer on CS substrate

CORROSION ALLOWANCE:

- Standard: 1.5 mm – 3.0 mm (per service life calculation)
- Aggressive media: 3.0 mm – 6.0 mm or clad construction

TECHNICAL SPECIFICATIONS — OVERVIEW

HEAT EXCHANGER COMPARISON TABLE

Parameter	Shell & Tube	Plate	Spiral
Heat Transfer Area	1 – 10,000+ m^2	0.5 – 2,500	10 – 2,000
Design Pressure	vac – 400 bar	up to 100	vac – 40
Design Temperature	-196 to +600°C	-196 to +400	-50 to +400
Min. Approach Temp (ΔT)	5 – 10°C	1 – 3°C	3 – 5°C
Fouling Resistance	Customizable	Low	Self-cleaning
Maintenance Access	Excellent	Good	Good
Footprint	Large	Compact	Compact
Scalability	Modular	Add plates	Custom

AIR COOLED SPECIFICATIONS:

Parameter	Range / Options
Heat Duty	0.1 – 100+ MW per unit
Process Pressure	Vacuum – 100 bar
Process Temperature	-50°C to +400°C
Air Flow Rate	10,000 – 2,000,000 m^3/h
Fan Power	5 – 500 kW per bay
Noise Level	<85 dB(A) @ 1m (with silencers)
Ambient Design	-40°C to +55°C (custom)

STANDARDS & COMPLIANCE

DESIGN & MANUFACTURING STANDARDS:

Standard	Description
TEMA Class R/C/B	Tubular Exchanger Manufacturers Assoc.
ASME Section VIII	Pressure Vessel Code, Div.1 / Div.2
ASME Section II	Material specifications
ASME B31.1 / B31.3	Power Piping / Process Piping
API 660 / 661 / 662	Shell & Tube / Air Cooled / Plate
EN 13445	Unfired pressure vessels (Europe)
EN 13458	Cryogenic vessels
GOST R 53677 / GOST R 52857	Russian standards for heat exchangers
PED 2014/68/EU	Pressure Equipment Directive
AD 2000 Merkblatt	German pressure vessel code
CRN (Canada)	Provincial registration

MATERIAL STANDARDS:

- ASTM A240 / EN 10028-7 — Stainless steel plates
- ASTM A213 / A268 — Stainless/titanium tubes
- ASTM A516 / A537 — Carbon steel plates
- ASTM B265 / B861 — Titanium tubes/sheets
- ASTM B443 / B446 — Inconel plaloy plates
- NACE MR0175 / ISO 15156 — Materials for sour service

APPLICATIONS BY INDUSTRY

OIL & GAS REFINING:

- **Crude preheat trains** — Shell & Tube, high fouling design
- **Hydrocracker / Hydrotreater effluent** — high pressure, H_2 service
- **FCCU main fractionator overhead** — condensation, corrosion control
- **Sulfur recovery (Claus)** — acid gas cooling, corrosion-resistant materials
- **Amine regeneration** — lean/rich amine exchangers, **SS 316/Duplex**

POWER GENERATION:

- **Thermal power** — HP/LP feedwater heaters, turbine extraction
- **Combined cycle** — HRSG economizers, evaporators, superheaters
- **Nuclear** — Component cooling water, essential service water (special grades)
- **District heating** — Primary/secondary network exchangers, large plate units

PETROCHEMICAL & CHEMICAL:

- **Ethylene plant** — cracked gas quench, propylene refrigeration
- **Ammonia / Methanol** — synthesis gas cooling, high pressure
- **Chlor-alkali** — chlorine cooling, caustic heating (Ti, Ni alloys)
- **Polymer production** — reactor cooling, monomer condensation
- **Acid production** — sulfuric, nitric, phosphoric acid coolers (Hastelloy, PTFE-lined)

ENGINEERING SUPPORT & SELECTION PROCESS

OUR ENGINEERING WORKFLOW:

1. DATA COLLECTION

- Process flow diagram (PFD) / P&ID
- Heat & mass balance, stream properties
- Operating pressure / temperature (normal & design)
- Media composition, fouling factors, corrosion data
- Space constraints, orientation, nozzle preferences
- Applicable standards, certification requirements

2. PRELIMINARY SELECTION

- Heat exchanger type recommendation
- Material selection per media & conditions
- Preliminary sizing (area, ΔP, approach temp)
- Configuration options (passes, baffle type, etc.)

3. DETAILED DESIGN (HTRI / Aspen EDR)

- Thermal rating & hydraulic calculation
- Mechanical design per ASME / EN / PED
- Stress analysis, nozzle loads, support design
- Vibration analysis (for shell & tube, air cooled)

4. DOCUMENTATION & APPROVAL

- Data sheet & specification
- General arrangement drawing (PDF/DWG)
- Calculation report (thermal, mechanical)
- Material specification & inspection test plan
- Commercial offer with lead time

WHAT WE DELIVER:

- ✔ Technical proposal with thermal summary & key parameters
- ✔ General arrangement drawing with dimensions & nozzle schedule
- ✔ Material specification (ASTM/EN/GOST equivalents)
- ✔ Quality plan: inspection points, NDT requirements, documentation
- ✔ Commercial offer: pricing, payment terms, delivery schedule

QUALITY CONTROL & DOCUMENTATION PACKAGE

MANUFACTURER QUALIFICATIONS:

- ✔ ISO 9001:2015 — Quality Management System
- ✔ ISO 14001:2015 — Environmental Management
- ✔ ISO 45001 — Occupational Health & Safety
- ✔ Welding Procedure Qualification (WPQR / WPS per ASME IX)
- ✔ NDT Personnel Certification (ASNT Level II/III)
- ✔ Authorized Inspection Agency (AIA) partnership

PRODUCT CERTIFICATIONS:

- ✔ PED 2014/68/EU — Category I to IV (as applicable)
- ✔ ASME Section VIII Div.1 / Div.2 U Stamp (on request)
- ✔ TEMA Class R/C/B compliance certificate
- ✔ API 660 / 661 compliance (for oil & gas applications)
- ✔ GOST-R / EAC Certification (for CIS markets)
- ✔ CRN registration (Canada), DOSH (Malaysia) — on request

REFERENCE PROJECTS

INDUSTRIES & APPLICATIONS:

- Refinery crude preheat train — Middle East
- Ethylene plant quench system — Southeast Asia
- LNG seawater vaporizer — Asia Pacific
- Ammonia plant synthesis gas cooler — India
- Offshore platform lube oil cooler — North Sea
- District heating plate exchangers — Eastern Europe

MATERIALS SUPPLIED:

- Stainless Steel 304/316/321 – 60% of projects
- Duplex/Super Duplex – 15% (offshore, chloride service)
- Titanium – 10% (seawater, chemical resistance)

TYPICAL PROJECT SPECIFICATIONS:

- Shell & Tube: DN800 shell, SS 321 tubes, 1,200 m^2, 40 bar, 350°C, TEMA BEU
- Plate (Welded): Titanium Gr.2, 850 m^2, seawater cooling, 10 bar, 80°C
- Spiral: Duplex 2205, 450 m^2, slurry service, self-cleaning design
- Air Cooled: 12-bay fin-fan, CS tubes/Al fins, 25 MW duty, desert ambient

WHY CHOOSE CORROTECH FOR HEAT EXCHANGERS?

DIRECT MANUFACTURER ACCESS

- No middlemen — competitive pricing, faster lead times

ENGINEERING EXPERTISE

- HTRI/Aspen EDR calculations, TEMA/ASME compliance

MATERIAL FLEXIBILITY

- CS, SS, Duplex, Titanium, Inconel, Hastelloy — all available

FULL DOCUMENTATION

- MTC 3.1, PED, ASME, GOST — ready for customs & audits

QUALITY ASSURANCE

- Hydro test, NDT, third-party inspection available

CONTACT US

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CONTACT INFORMATION

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Website:
https://pipefittings-hg.com/

Business Hours: Monday – Friday: 9:00 – 18:00 (HKT)
Weekend: By appointment

REQUEST FOR QUOTATION:

- Type of heat exchanger (Shell & Tube / Plate / Spiral / Air Cooled)
- Process data: flow rates, inlet/outlet temperatures, pressures
- Media composition (both sides), fouling factors
- Material preferences or corrosion data
- Applicable standards (TEMA/ASME/API/EN/GOST)
- Space constraints, nozzle orientation preferences
- Delivery destination and required date

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